BEIR: A Heterogeneous Benchmark for Zero-Shot Evaluation of Information Retrieval Models

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***Abstract** — Dense retrieval models trained on MS-MARCO have set new state-of-the-art benchmarks on in-domain evaluation. However, their generalization ability across heterogeneous out-of-domain search tasks remains questioned. We introduce BEIR, a diverse benchmark spanning 18 retrieval datasets across 9 domains (bio-medical, finance, legal, tweet retrieval, and technical QA). Our empirical evaluation reveals that while DPR models degrade significantly under zero-shot domain shifts, classical BM25 remains a formidable baseline. Furthermore, we demonstrate that a two-stage pipeline combining Hybrid Sparse-Dense retrieval with Cross-Encoder reranking achieves superior NDCG@10 (+14.2%) across all BEIR domains, though at the expense of increased per-query inference latency.*

1. Introduction

Neural information retrieval models have achieved impressive breakthroughs on large datasets like MS-MARCO. Despite these gains, real-world deployment requires zero-shot robustness across specialized domains where fine-tuning data is unavailable. We construct BEIR to evaluate whether dense dual-encoders genuinely surpass BM25 when tested out-of-domain.

2. Methodology: Hybrid Retrieval & Cross-Encoder Reranking

We formulate a multi-stage retrieval architecture. In the first stage, we compute Reciprocal Rank Fusion (RRF) combining sparse BM25 scores and dense MiniLM embeddings: RRF_score(d) = sum(1 / (60 + rank_i(d))). In the second stage, the top-100 candidates are rescored using a Cross-Encoder (MiniLM-L-6-v2) that attends simultaneously to the concatenated query and passage tokens: CrossEncoder(q, p) = Softmax(W · BERT(q [SEP] p)).

3. Experimental Evaluation on BEIR

We evaluate 17 retrieval architectures across 18 datasets including BioASQ, COVID-19, FiQA, CQADupStack, and Touche-2020. Metrics include NDCG@10, Mean Reciprocal Rank (MRR@10), and wall-clock query latency (ms).

4. Results & Disagreements with DPR

Contradicting earlier claims that dense retrieval unconditionally supersedes sparse search, BM25 outperforms standalone DPR on 11 of the 18 zero-shot BEIR datasets. Specifically, on biomedical datasets (BioASQ), BM25 achieves 0.654 NDCG@10 whereas DPR achieves only 0.388 NDCG@10. However, our Hybrid + Cross-Encoder model achieves 0.768 NDCG@10, setting the state-of-the-art across all benchmarks.

5. Computational Limitations & Latency Overhead

A major drawback of Cross-Encoder reranking is inference latency. Rescoring 100 passages with MiniLM Cross-Encoder adds 42ms to 85ms per query on GPU (and over 350ms on CPU). For high-throughput real-time systems, this computational overhead represents a critical deployment bottleneck.

6. Conclusion & Research Gaps

Zero-shot generalization requires hybrid sparse-dense indexing and cross-attention reranking. We leave for future work the investigation of compressed cross-encoders, late-interaction models (e.g. ColBERT), and adaptive routing algorithms.